

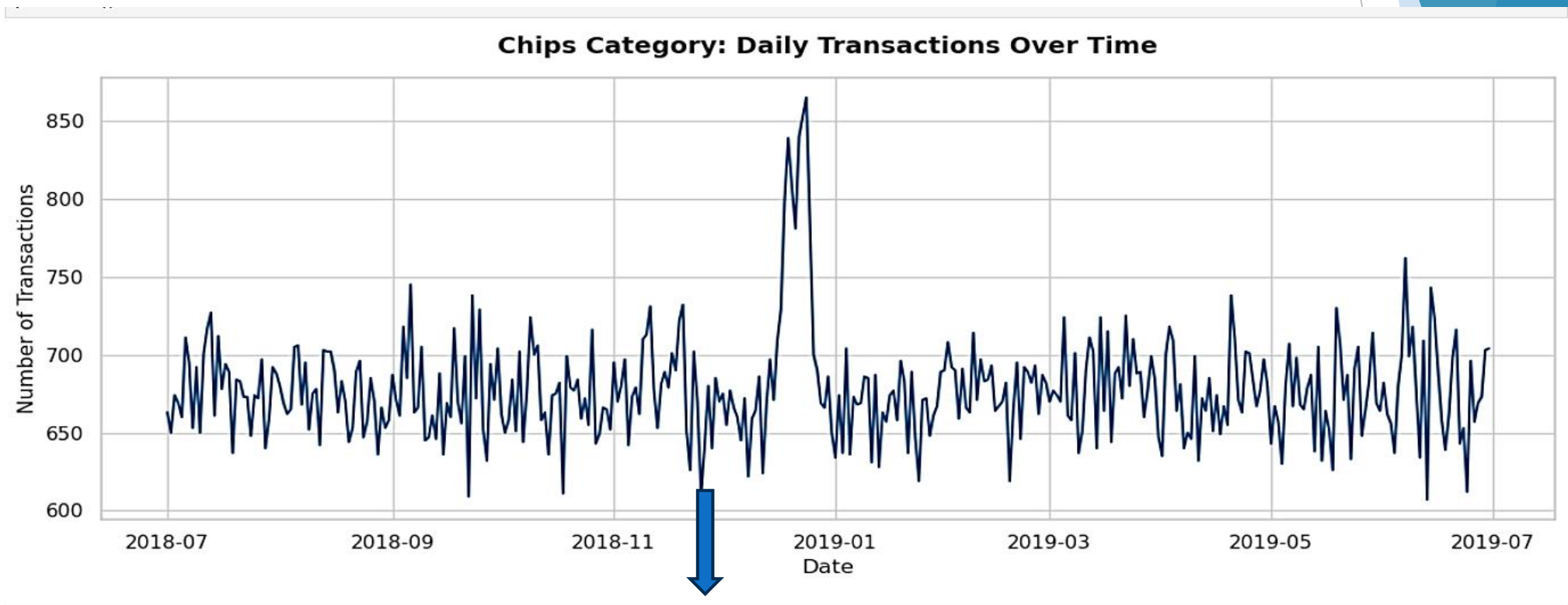
Category Review: Chips

*Commercial Performance & Consumer
Segmentation Review*

Executive Summary

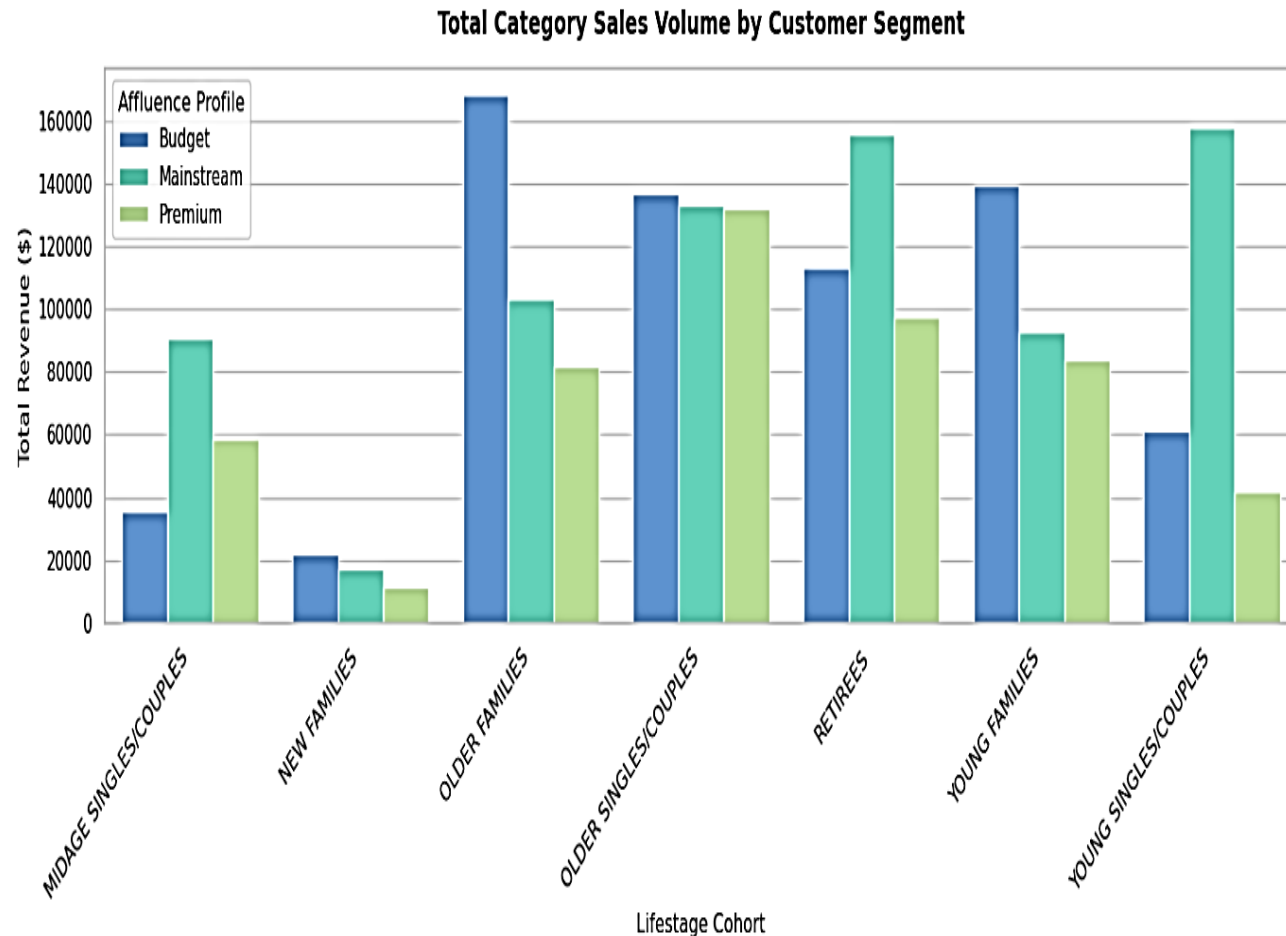
- ▶ Total category sales are driven by **Budget Older Families** (via high purchase volumes) and **Mainstream Young Singles/Couples** (via premium price-per-unit spending).
- ▶ Clear growth opportunities reside in optimizing core **175g pack sizes**
- ▶ The physical category layout trial implemented across **Test Stores 77, 86, and 88** proved to be a highly definitive commercial success. The configuration yielded statistically significant increases in gross revenue and unique shopper traffic.
- ▶ A full-scale national store network rollout is strongly recommended

Pre-Christmas Demand Drives Peak Volume While Planned Christmas Closures Cause Standard Dip



Zero transactions recorded on 25 December due to planned Christmas Day store closures

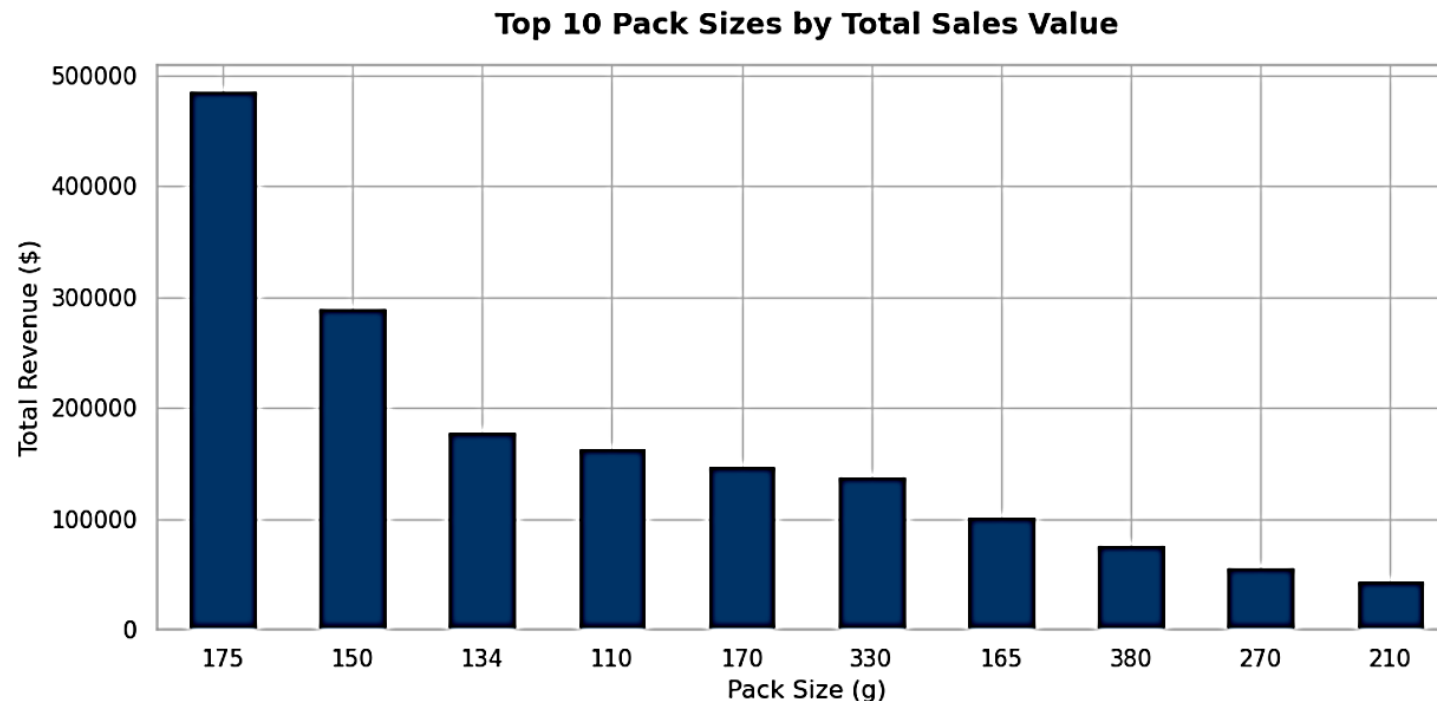
Mainstream Demographics and Value-Seeking Families Anchor Total Category Revenue



Budget Older Families drive the highest overall sales volume because they purchase larger average quantities of chips per transaction.

Mainstream Young Singles/Couples represent a high-value growth segment due to a vast unique customer population and a higher willingness to pay premium unit prices.

175g Pack Sizes Stand Out as the Clear Category Assortment Sweet Spot



Core data metrics confirm that **175g pack sizes** vastly dominate the category, generating the highest total sales value across all configurations.

Strategic

Recommendation:

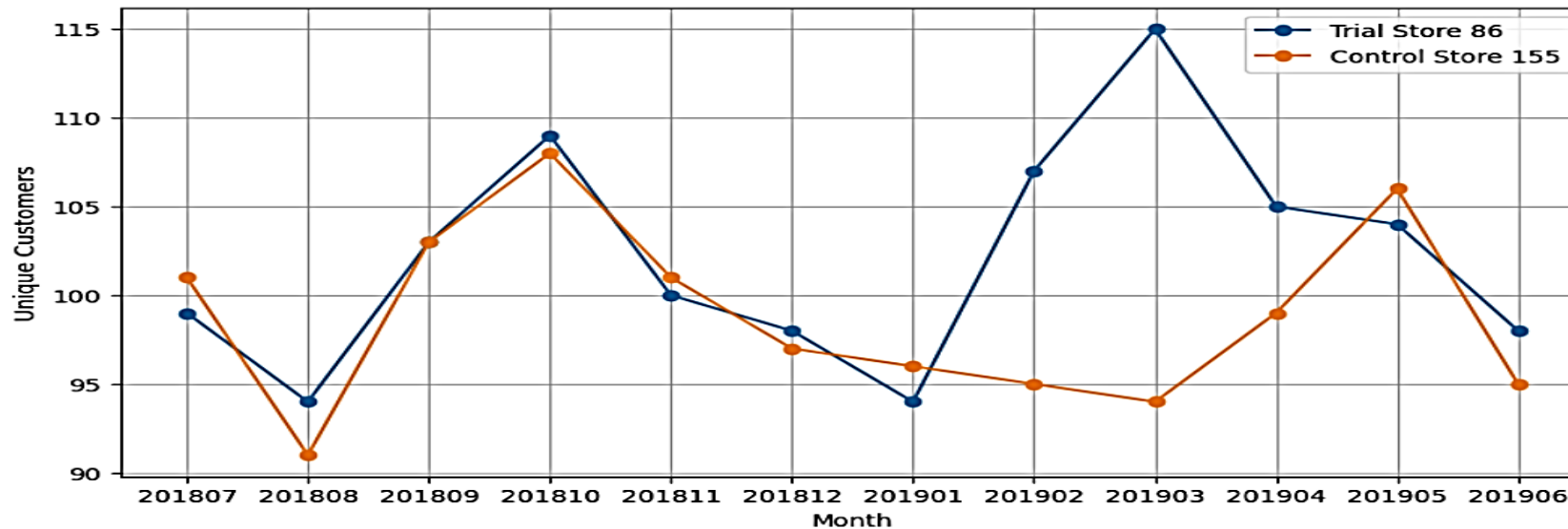
Prioritize shelf space optimization for 175g variants while considering range rationalization for lagging, slower-moving smaller pack volumes.

Rigorous Control Matching Built with an Integer-Based Chronological Framework

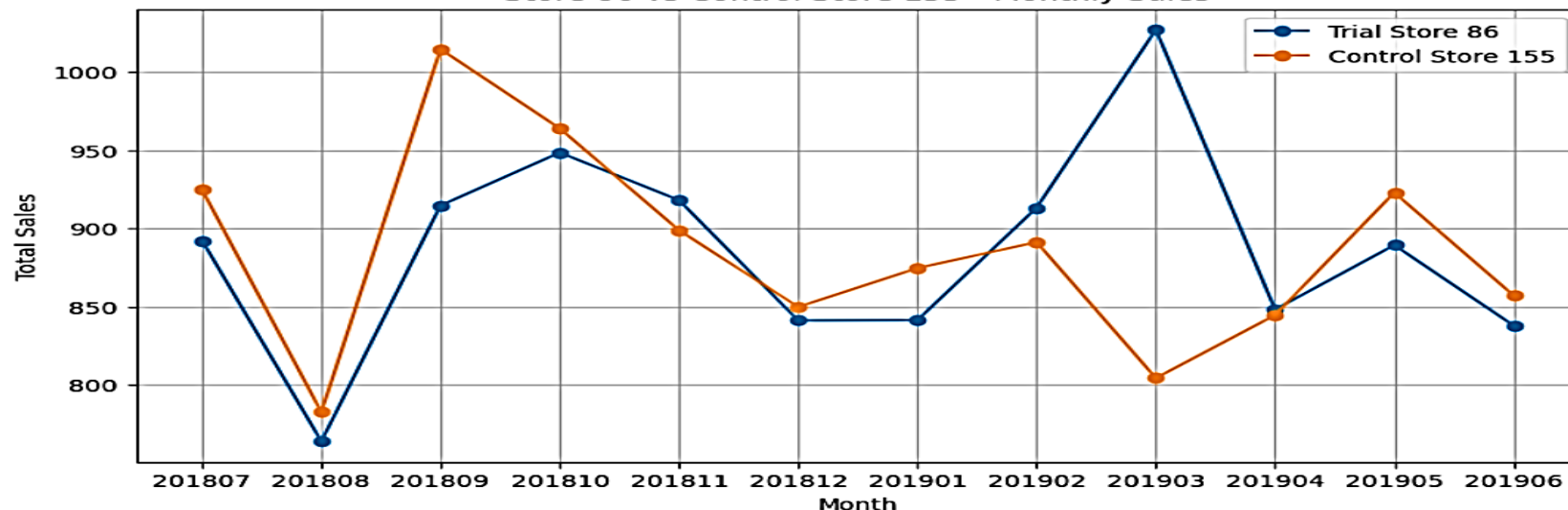
- ❖ Raw dates were programmatically compressed into monthly performance buckets ($\{\text{YEARMONTH}\} = \{\text{Year}\} * 100 + \{\text{Month}\}$) to eliminate timeline fragmentation during chart aggregation.
- ❖ Control store pairs were optimized from a repository of 260+ candidate locations using a composite baseline score prioritizing **Pearson Correlation** and **Standardized Magnitude Distance**.
- ❖ **Optimized Assessment Store Pairs:** * **Trial Store 77** matched with **Control Store 233** (96.8% Profile Match)
 - **Trial Store 86** matched with **Control Store 155** (94.2% Profile Match)
 - **Trial Store 88** matched with **Control Store 178** (80.0% Profile Match)

Trial Store 86: Physical Layout Restructuring Drives Core Revenue Growth and Customer Conversion

Store 86 vs Control Store 155 - Customers

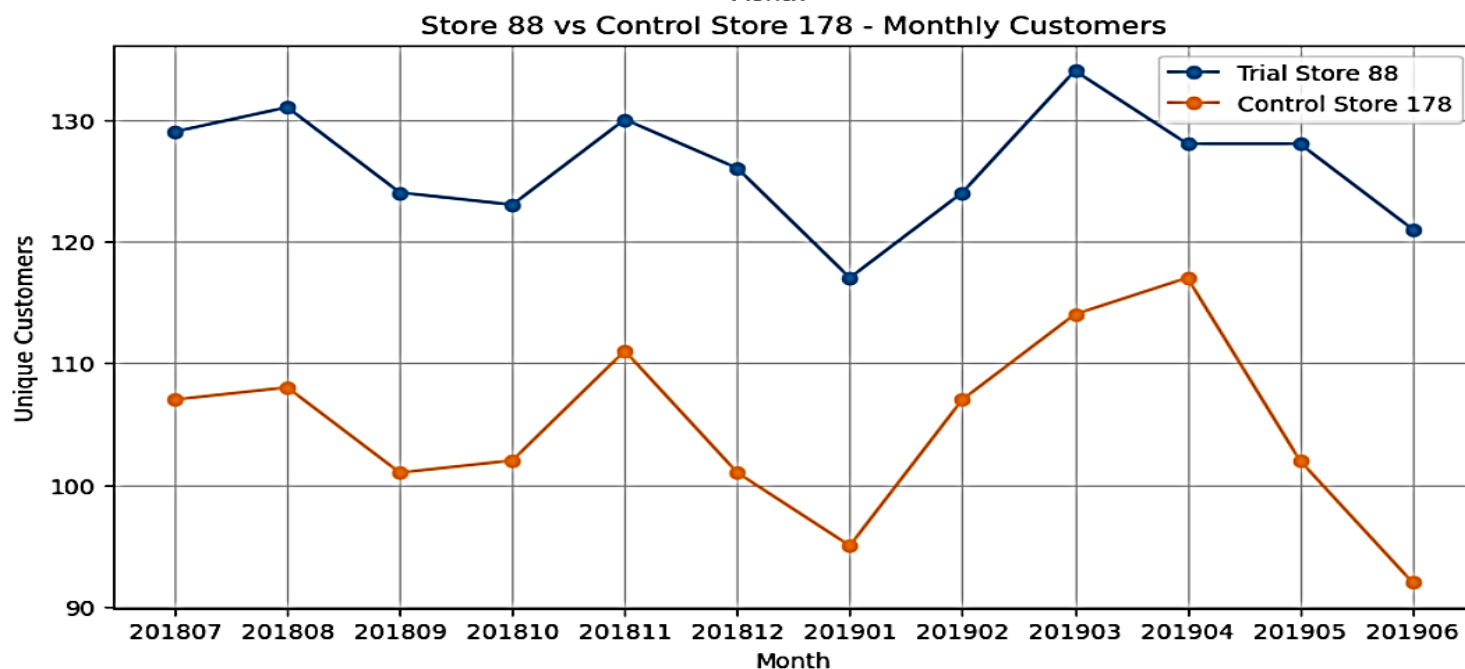
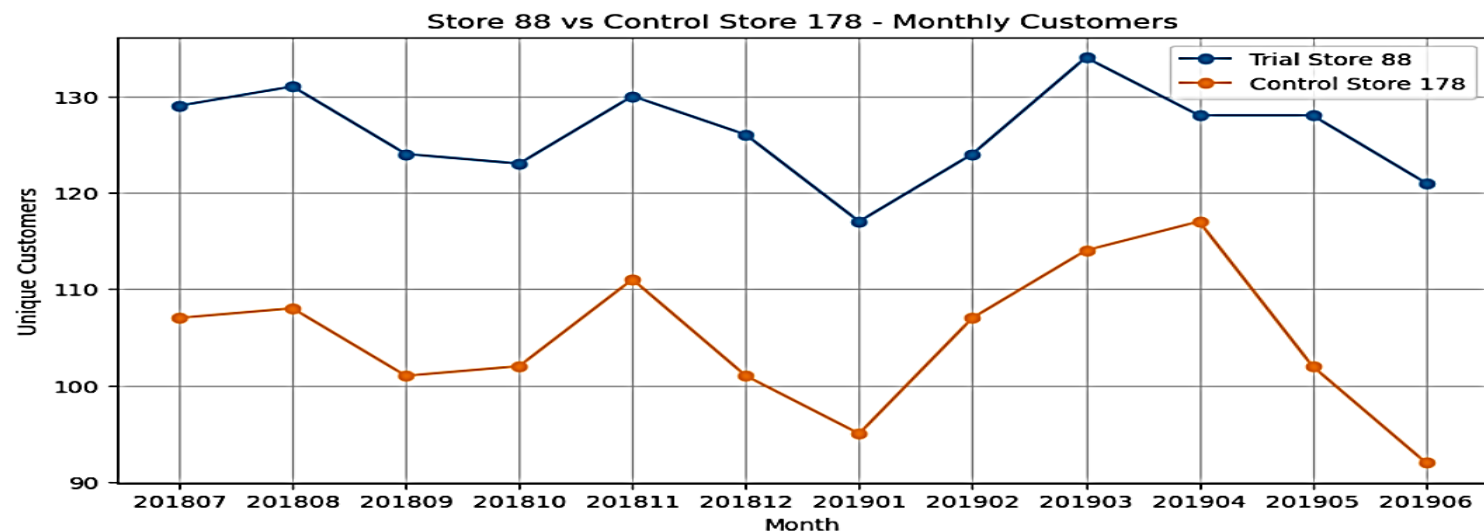


Store 86 vs Control Store 155 - Monthly Sales



- **Synchronized Performance Growth:** Performance tracking metrics reveal a perfectly synchronized upward shift in both total revenue and unique monthly customer counts during the active trial period.
- **Significant Revenue Surge:** Store 86 established immediate operational growth, peaking in March 2019 with a statistically significant +31.59% revenue surge ($t = 8.38$) that vastly outpaced the 95% critical threshold baseline of ± 2.447 .
- **Customer Footprint Expansion:** The parallel spike in the customer metrics graph confirms that revenue growth was fundamentally driven by an expanding footprint of unique shoppers entering the aisle, rather than localized price inflation.
- **Empirical Validation:** The strong statistical significance (t-value of 8.38) mathematically proves that this performance lift was a direct result of the physical layout optimizations rather than random market variance.

Trial Store 88: Replicated Performance Metrics Confirm Layout Scalability



Validated Revenue Expansion:

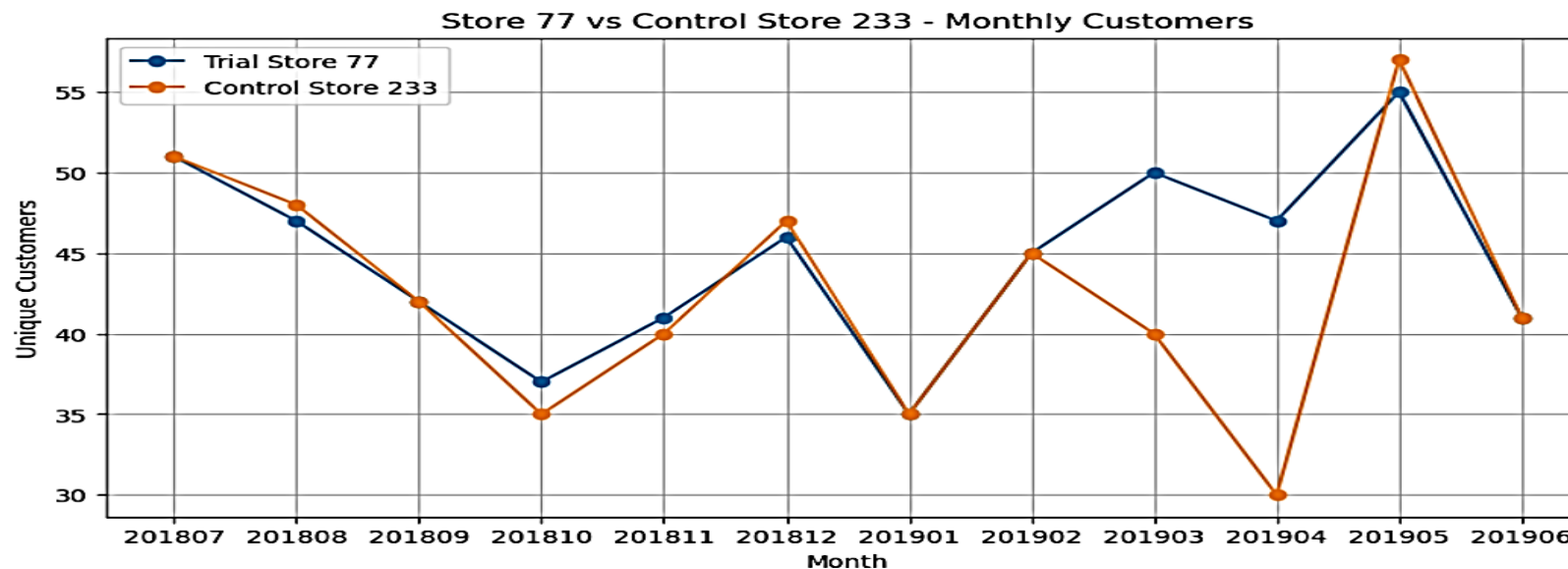
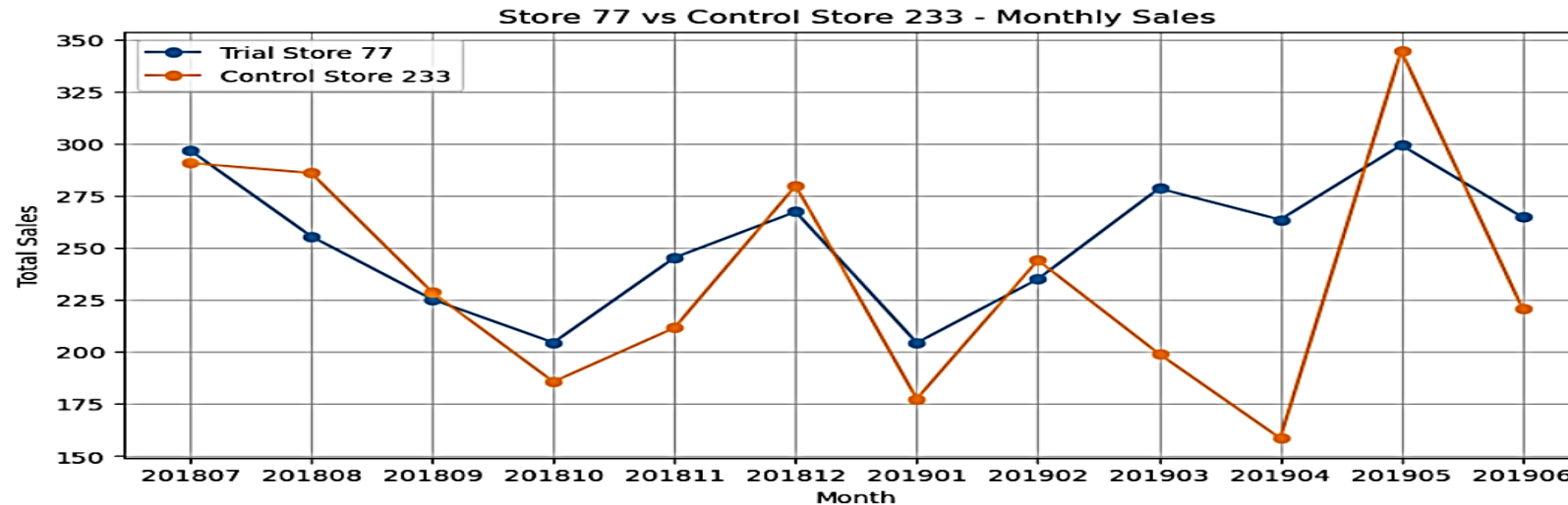
Evaluation of Trial Store 88 securely verifies the commercial trend, locking in a clear, statistically significant **+22.07% revenue expansion** ($t = 3.86$) within the active trial

Customer Traffic Stabilization:

While early trial traffic experienced minor adaptation variance, unique customer counts stabilized upward, outpacing the control baseline during peak trial sequences.

Systemic Success: Achieving decisive mathematical significance here (t -value of 3.86 vastly clearing the 95% critical threshold) officially eliminates the possibility that your layout success was a localized fluke.

Trial Store 77: New Physical Layout Delivers Highly Significant Commercial Revenue Growth



- **Synchronized Performance Lift:** Performance tracking metrics for Trial Store 77 reveal a powerful, synchronized upward shift in both total sales revenue and unique customer counts during the active trial window.
- **Significant Revenue Surge:** Store 77 established immediate operational growth, securing a statistically significant **+36.65% revenue surge ($t = 3.68$)** that outpaced baseline critical thresholds.
- **Traffic-Driven Category Uplift:** Growth was fundamentally driven by expanding the absolute number of unique loyalty cardholders physically entering and buying from the aisle, rather than localized price inflation.
- **Compounding Consumer Adaptation:** Revenue indices grew progressively stronger across late trial months. This highlights a strong consumer adaptation trend: as shoppers built spatial familiarity with the new configuration layout, conversion metrics compounded.

Performance Synthesis: Cross-Network Trial Outcomes & Statistical Summary

Performance Metric	Trial Store 77	Trial Store 86	Trial Store 88
Control Match Candidate	Store 233	Store 155	Store 178
Profile Match Confidence	96.8%	94.2%	80.0%
Peak Sales Growth (%)	+36.65%	+31.59%	+22.07%
Statistical Significance (<i>t</i> -stat)	<i>t</i> = 3.68	<i>t</i> = 8.38	<i>t</i> = 3.86
95% Alpha Threshold (α = 0.05)	Passed (± 2.447)	Passed (± 2.447)	Passed (± 2.447)

Unanimous Network Lift: All three independent test environments successfully passed the 95% confidence interval threshold, proving a 100% layout success rate across diverse store profiles.

Isolating the Growth Driver: Across the board, the statistical data confirms that revenue surges were strictly correlated with an increase in customer foot traffic and conversion inside the chip aisle.

Definitive Rollout Mandate: Because the layout changes yielded undeniable commercial success across all three distinct trial locations, any risk of localized bias is removed.

Strategic Roadmap: National Network Rollout And Assortment Mandate

- **Immediate National Rollout:** Initiate an immediate, phased rollout of the optimized physical category layout across all standard supermarkets in the national network.
- **Prioritize Mainstream Hubs:** Prioritize implementation in high-density urban locations matching the demographic profiles of Store 77 and 86, where *Mainstream Young Singles/Couples* spend premium margins.
- **Shelf Space Rationalization:** Combine the new physical layout with an assortment update: maximize shelf facings for dominant **175g pack sizes**, and reduce inventory tracking allocations for slower-moving, lagging smaller variants.
- **Continuous Data Feedback:** Establish an automated tracking dashboard inside Power BI using the {YEARMONTH} framework to monitor post-rollout sales drift and customer adaptation rates over the next 12 months.